

Lancer1911 TTS Offline

Fully Offline Multi-Voice Text-to-Speech Synthesis

完全离线多角色情绪可控文本转语音合成工具

Installation & User Guide v0.3a · 安装与使用说明

1. System Requirements 系统要求

Item 项目	Requirement 要求
Hardware 硬件	Apple Silicon Mac (M1 / M2 / M3 / M4 or later) Apple Silicon Mac (M1 / M2 / M3 / M4 及以上)
Memory 内存	16 GB minimum, 32 GB recommended 最低 16 GB，推荐 32 GB
macOS 系统	macOS 13 Ventura or later; 14 Sonoma or later recommended macOS 13 Ventura 或更高，推荐 macOS 14 Sonoma 及以上
Storage 存储空间	~2–10 GB for TTS models (varies by variant) TTS 模型约需 2–10 GB (因模型而异)
Internet 网络	Only needed for first-time model download 仅首次下载模型需要联网
Text formats 文本格式	TXT, Markdown, DOCX, SRT, PDF, EPUB 支持 TXT、Markdown、DOCX、SRT、PDF、EPUB

⚠️ Lancer1911 TTS Offline runs entirely on-device. Your text and synthesized audio never leave your Mac. ⚠️ Lancer1911 TTS Offline 完全在本地运行，文本和合成音频不会上传到任何服务器。

💡 TTS Offline requires far less memory than ASR Offline. A 16 GB M1 Mac can run the default 1.7B model comfortably. 💡 TTS Offline 的内存需求远低于 ASR Offline，16 GB M1 Mac 可流畅运行默认 1.7B 模型。

2. Installation 安装步骤

2.1 Quick Install (Recommended) 一键安装脚本（推荐）

The included install.sh handles environment setup, package installation, and model download automatically.

内附的 install.sh 脚本会自动完成环境配置、依赖安装和模型下载。

```
# Standard install (recommended) bash install.sh # Skip model download (already downloaded) bash install.sh --no-model # Use China mirror (hf-mirror.com) bash install.sh --cn # Combine options bash install.sh --no-model --cn
```

2.2 Manual Installation 手动安装

2.2.1 Install Homebrew 安装 Homebrew（已安装可跳过）

Open Terminal and run: 打开终端，运行以下命令：

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

2.2.2 Install ffmpeg 安装 ffmpeg

Required for MP3 output. WAV output works without ffmpeg.

MP3 输出需要 ffmpeg；WAV 输出无需此工具。

```
brew install ffmpeg
```

2.2.3 Set Up Python Environment 配置 Python 虚拟环境

Run these commands once in Terminal: 在终端依次运行以下命令（只需执行一次）：

```
# Create virtual environment python3 -m venv ~/tts-offline-env # Activate it source ~/tts-offline-env/bin/activate # Install dependencies pip install -r requirements.txt
```

If you use the packaged .app, the app is lightweight and relies on the local Python environment and downloaded models.

如果使用打包后的 .app，应用本体较小，仍依赖本地 Python 环境和已下载模型。

3. Download TTS Models 下载 TTS 模型

Lancer1911 TTS Offline uses Qwen3-TTS models from the mlx-community organization.

Download once; they are cached locally and work offline.

Lancer1911 TTS Offline 使用 mlx-community 的 Qwen3-TTS 模型。下载一次后缓存在本地，之后可完全离线使用。

TTS Model Family TTS 模型族

Model 模型	Size 大小	Voices 内置音色	Recommended for 适用场景
1.7B-CustomVoice-8bit	~1.7 GB	9 built-in 9 种内置	Default; best for most users 默认推荐，适合大多数用户
0.6B-CustomVoice-8bit	~0.7 GB	9 built-in 9 种内置	Low memory / faster synthesis 内存有限或速度优先
1.7B-Base-8bit	~1.7 GB	None (cloning only) 无（仅克隆）	Voice cloning from reference audio 声音克隆，需提供参考音频

Model 模型	Size 大小	Voices 内置音色	Recommended for 适用场景
1.7B-VoiceDesign-8bit	~1.7 GB	Text-described 文字描述	Describe a voice in text 用文字描述想要的音色

```
source ~/tts-offline-env/bin/activate # Recommended default - 9 built-in voices,
emotion instructions hf download mlx-community/Qwen3-TTS-12Hz-1.7B-CustomVoice-8bit
# Lightweight alternative # hf download mlx-community/Qwen3-TTS-12Hz-0.6B-
CustomVoice-8bit # Voice cloning (requires Base model) # hf download mlx-
community/Qwen3-TTS-12Hz-1.7B-Base-8bit
```

If downloads are slow or failing, use the China mirror: 下载缓慢或失败时, 可使用镜像:

```
# Add before hf download command: HF_ENDPOINT=https://hf-mirror.com hf download
mlx-community/Qwen3-TTS-12Hz-1.7B-CustomVoice-8bit
```

💡 If 'hf' command is not found: pip install -U huggingface_hub 💡 若提示找不到 hf 命令: pip install -U huggingface_hub

4. Install Lancer1911 TTS Offline.app 安装应用

4.1 Copy to Applications 拷贝到应用程序文件夹

Open the DMG file and drag Lancer1911 TTS Offline into the Applications folder.

打开 DMG 文件, 将 Lancer1911 TTS Offline 拖入 Applications (应用程序) 文件夹。

4.2 Bypass Gatekeeper 绕过安全限制

Because Lancer1911 TTS Offline is not notarized by Apple, macOS may block it on first launch.

由于 Lancer1911 TTS Offline 未经 Apple 公证, 首次打开可能被系统拦截。

- Option A — Right-click Lancer1911 TTS Offline in Applications → Open → click Open. 方法 A — 在 Applications 中右键 Lancer1911 TTS Offline → 打开 → 弹窗中点击「打开」。
- Option B — Terminal: 方法 B — 终端命令:

```
xattr -cr "/Applications/Lancer1911 TTS Offline.app"
```

5. Using Lancer1911 TTS Offline 使用说明

5.1 First Launch 首次启动

Allow 10–30 seconds on first launch for the TTS model to load into memory. The status bar shows Ready when the app is fully loaded.

首次启动需要 10–30 秒加载 TTS 模型, 状态栏显示「就绪」后即可使用。

⚠️ If no model is detected, the app shows a Download Wizard automatically. Follow the on-screen instructions. ⚠️ 若未检测到模型, 应用会自动显示下载向导, 按提示操作即可。

5.2 Plain Text Synthesis 普通文本合成

Step 步骤	Action 操作
1	Select a voice in the Voice tab (left sidebar). 在左侧音色标签页选择音色。
2	Paste or type text in the editor, or drag-and-drop a TXT / MD / DOCX / PDF / EPUB file. 在编辑框粘贴文本，或拖放文件（TXT / MD / DOCX / PDF / EPUB）。
3	Adjust speed with the Speed slider. 使用语速滑块调整语速。
4	Click  Synthesize. 点击  开始合成。
5	After synthesis, the playback bar appears. Click  Download to save the audio. 合成完成后，播放栏出现。点击  下载音频保存。

5.3 Multi-Character Audiobook (Dialog Table) 多角色有声书（对话表格）

Dialog Table mode lets you produce full audiobooks or drama scripts with per-line voice, emotion, and direction control.

对话表格模式支持逐句设置音色、情绪和导演指令，可制作完整有声书或广播剧。

Step 步骤	Action 操作
1	Click Dialog Table to switch to the three-column table view. 点击「三列表格」切换到对话表格视图。
2	Each row = one line of speech: Speaker / Instruction / Text. 每行代表一段语音：发言人 / 指令 / 文本。
3	Add rows with + Row, or import a CSV script with Load CSV. 点击「+ 行」添加行，或用「加载 CSV」导入剧本。
4	Click  Synthesize. Lines are synthesized in order and concatenated. 点击  开始合成，各行按顺序合成并拼接为单个音频。
5	After synthesis, click any row to jump to its position (reverse follow). 合成后，点击任意行跳转到对应播放位置（反向跟随）。

CSV format (header optional): CSV 格式（表头可选）：

```
Speaker,Instruction,Text
Serena,,Once upon a time in a quiet village...
Ryan,whispering urgently,We need to leave. Now. Vivian,warm and
reassuring,Everything will be fine.
```

5.4 Voice Cloning 声音克隆

Step 步骤	Action 操作
1	Switch to the Clone tab. 切换到「克隆」标签页。

Step 步骤	Action 操作
2	Enter a Clone Name and optional Reference Text (what is said in the audio). 输入克隆名称，可选填写参考音频文字（音频中说的内容）。
3	Choose a Base Voice (style reference). 选择基础音色（风格参考）。
4	Upload WAV or MP3 reference audio (5–30 s, clean recording recommended). 上传 WAV 或 MP3 参考音频（5–30 秒，建议清晰无噪音）。
5	Click Start Clone. The cloned voice appears in the voice list. 点击「开始克隆」，克隆音色将出现在音色列表中。

⚠ Voice cloning requires a Base model (e.g. Qwen3-TTS-12Hz-1.7B-Base-8bit). Switch to a Base model in the Model tab and click Load Model first. ⚠ 声音克隆需要 Base 模型（如 Qwen3-TTS-12Hz-1.7B-Base-8bit）。请先在模型标签页切换到 Base 模型并点击「加载模型」。

5.5 Interface Overview 界面说明

Area 区域	Function 功能
Top bar 顶栏	Status, Save, Load, Advanced, Language switch, Theme toggle 状态、保存、加载、高级设置、语言切换、主题切换
Voice tab 音色标签页	Speed slider, voice grid (built-in + cloned voices) 语速滑块、音色网格（内置音色 + 已克隆音色）
Model tab 模型标签页	Model selector, Load Model, output format, output directory 模型选择、加载模型、输出格式、输出目录
Clone tab 克隆标签页	Reference audio upload, clone name, base voice, Start Clone 参考音频上传、克隆名称、基础音色、开始克隆
Text editor 文本编辑区	Plain text input, file upload, dialog table 普通文本输入、文件上传、对话表格
Playback bar 播放栏	Play/pause, seek, follow, reverse follow, download, subtitles 播放/暂停、进度拖动、跟随、反向跟随、下载、字幕

5.6 Built-in Voices 内置音色

Voice 音色	Gender 性别	Language 语言	Style 风格
Serena	Female 女	zh/en	Warm, professional 温暖、专业
Vivian	Female 女	zh/en	Bright, expressive 明快、富有表现力
Uncle Fu	Male 男	zh/en	Deep, authoritative 低沉、威严
Ryan	Male 男	zh/en	Calm, clear 沉稳、清晰

Voice 音色	Gender 性别	Language 语言	Style 风格
Alden	Male 男	zh/en	Casual, conversational 随意、对话感
Ono Anna	Female 女	zh/en	Gentle, measured 温柔、从容
Sohee	Female 女	zh/en	Energetic, youthful 活力、青春
Eric	Male 男	zh/en	Neutral, versatile 中性、多用途
Dylan	Male 男	zh/en	Rich, narrative 醇厚、叙事感

💡 All built-in voices support Chinese and English and respond to emotion/style instructions. 💡
所有内置音色支持中英文，并响应情绪 / 风格指令。

5.7 Emotion & Style Instructions 情绪与风格指令

The Control Instruction field accepts free-form natural language. Examples:
指令字段接受自由格式自然语言。示例：

Instruction 指令	Effect 效果
Speak slowly and clearly 语速缓慢而清晰	Measured pace, enunciation 有节奏感，发音清晰
Whispering, tense 低声耳语，紧张	Hushed delivery 轻声但带有张力
Cheerful and upbeat 轻快活泼	Lighter, faster cadence 语速稍快，情绪明亮
Calm but slightly worried 平静但略带担忧	Restrained affect with tension 克制情感，底色有忧虑
Authoritative, formal 威严、正式	Deliberate, composed 从容、庄重
Melancholy, distant 忧郁，若有所失	Slower, softer, trailing 缓慢、轻柔，有尾音
Warm storytelling tone 温暖的讲故事语气	Relaxed narrative register 放松的叙事腔调

5.8 Settings 设置说明

Advanced TTS Parameters 高级 TTS 参数

Click Advanced in the top bar to open the Advanced Settings panel.
点击顶栏「Advanced / 高级设置」打开高级参数面板。

Parameter 参数	Default 默认值	Guidance 说明
Temperature	0.9	Sampling randomness. Lower (0.6–0.8) for stable output; higher for expressive variation. 采样随机性。降低（0.6–0.8）可使输出更稳定；升高可增加表现变化。
Top-P	0.9	Nucleus sampling threshold. 核采样概率阈值。
Top-K	50	Number of candidate tokens. 候选 token 数量。
Max Tokens	4096	Max tokens per chunk. Increase for very long sentences. 每分段最大 token 数，超长句子可适当调大。
Chunk Size	250	Max characters per chunk. Reduce if synthesis is unstable. 每分段最大字符数，若合成不稳定可适当降低。
Skip Brackets	On	Ignores text inside () brackets (stage directions, annotations). 忽略括号内文字（舞台说明、标注）。
Numbers to Words	On	Converts Arabic numerals to spoken form before synthesis. 合成前将阿拉伯数字转换为中文读法。
Paragraph Gap (ms)	300	Silence between synthesized chunks. 各合成分段之间插入的静音时长。
Fade (ms)	10	Fade-in/out duration at audio edges. 音频首尾的淡化时长。
Pitch Shift	0	Pitch adjustment in semitones (0 = original). 音调调整（半音，0 = 原始音调）。

Built-in Presets 内置 Presets

Preset	Best for 适用场景
Audiobook (stable long text)	Long documents, chapter-level synthesis 长文档、章节级合成
Audiobook (mellow)	Deeper, more deliberate pacing 更沉稳的有声书节奏
Podcast (natural)	Conversational narration, varied cadence 对话感旁白，节奏自然
Dubbing (fast)	Short content, voice-over, faster delivery 短内容、配音
Reading (slow & clear)	Clear enunciation, educational content 发音清晰，适合教育内容

💡 Custom presets are saved to ~/.tts_offline_settings.json 💡 自定义 preset 保存在 ~/.tts_offline_settings.json

5.9 Session Files (.ttso) 会话文件

.ttso files save the complete session: script, voice selection, settings, and per-segment timestamps. Keep the .ttso file and the output audio in the same folder for automatic playback restoration.

.ttso 文件保存完整会话：剧本、音色选择、设置和逐段时间戳。将 .ttso 与输出音频保存在同一目录，可自动恢复播放。

```
| ProjectFolder/ |— my_audiobook.ttso |— my_audiobook.wav
```

5.10 Export 导出

Click the Subtitles menu to export timing data alongside the synthesized audio.

点击「字幕」菜单可导出时间戳数据与合成音频。

Format 格式	Description 说明
SRT 字幕文件	Standard subtitle format for video editors 标准字幕格式，可导入视频软件
TXT 纯文本	Plain text with timestamps 每条含时间戳的纯文本
Markdown	Suitable for Obsidian, Notion, documentation 适合 Obsidian、Notion 或文档工具
JSON 结构化数据	Full metadata: text, speaker, start_ms, end_ms 含文本、发言人、起止时间的完整结构化数据
Excel (CSV)	Tab-separated for spreadsheet import 制表符分隔格式，可导入电子表格

6. Troubleshooting 常见问题

Issue 问题	Solution 解决方法
Window shows "Connecting" for a long time 窗口长时间显示「连接中」	Wait 10–30 s for the model to load. Status bar shows Ready when done. 等待 10–30 秒，状态栏显示「就绪」后即可操作。
No models detected 未检测到模型	The app shows a Download Wizard automatically. Run: hf download mlx-community/Qwen3-TTS-12Hz-1.7B-CustomVoice-8bit If hf is not found: pip install -U huggingface_hub 应用会自动显示下载向导。若 hf 命令找不到：pip install -U huggingface_hub
Synthesis sounds unstable or cuts off 合成结果不稳定或被截断	Reduce Temperature (e.g. to 0.7) and Chunk Size (e.g. to 150). Add punctuation to long sentences. 降低 Temperature（如调至 0.7）和分段字数（如调至 150）；超长句子可添加标点引导分段。

Issue 问题	Solution 解决方法
Voice cloning does not work 声音克隆无效	Cloning requires a Base model. Switch to 1.7B-Base-8bit in the Model tab and click Load Model. 克隆需要 Base 模型。在模型标签页切换到 1.7B-Base-8bit 并点击「加载模型」。
MP3 output fails MP3 输出失败	Install ffmpeg: brew install ffmpeg. WAV output works without any extra dependency. 安装 ffmpeg: brew install ffmpeg。WAV 输出无需额外依赖。
CSV import fails CSV 导入失败	Ensure UTF-8 encoding and max 3 columns: Speaker, Instruction (optional), Text. 确保 UTF-8 编码，最多三列：发言人、指令（可选）、文本。
Session loads but audio is missing 会话加载后找不到音频	Keep the .tts and the audio file in the same folder with the original filename. 将 .tts 与音频保存在同一目录，且保持原始文件名。
Port already in use 端口已被占用	Default port is 17435. Check: lsof -i :17435 Kill: lsof -ti :17435 xargs kill -9 默认端口 17435。检查: lsof -i :17435 终止: lsof -ti :17435 xargs kill -9

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MLX · Qwen3-TTS · FastAPI · pywebview · ffmpeg · mlx-audio